Project Title:

Sales Data Cleaning and Analysis Using Excel

Story:

You have been hired as a Data Analyst intern by a retail company to help analyze its sales performance. The company has provided you with a raw sales dataset containing details of 1000 customer orders, but the data is **incomplete, inconsistent, and messy**.

Your task is to use **Microsoft Excel** to:

- **Clean the data**
- **Perform basic and advanced calculations**
- **Analyze sales performance using pivot tables and formulas**
- **Create visualizations and a summary dashboard**

This project is designed to strengthen your skills in Excel operations like data cleaning, formula writing, conditional formatting, data analysis, and chart building.

Objective:

Perform complete data analysis and cleaning on a real-world–like sales dataset using Excel tools only — **no Python or SQL required**.

Dataset Overview:

The dataset contains the following columns:

- Order ID
- Date
- Customer Name
- Region
- Product
- Category
- Quantity
- Price
- Discount
- Total *(with some incorrect/missing values)*

Tasks to be Performed:

🔹 1. Data Cleaning

- Identify and remove duplicate rows
- Fix formatting issues (e.g., lowercase to Proper Case)
- Handle missing values in:
  - Region
  - Customer Name
  - Price
- Convert discount from % string to number (optional)

🔹 2. Add Derived Column

- Recalculate the **Total** using the formula:
  `Total = Quantity * Price * (1 - Discount)`

  Use Excel formulas, not manually calculated values

🔹 3. Summary Calculations

- Total sales (sum of Total)
- Average order value
- Highest single order value
- Total number of orders
- Top 5 products by revenue

🔹 4. Conditional Formatting

- Highlight:
  - Orders with high discount (>15%)
  - Orders with Total > ₹10,000
  - Missing or blank fields

🔹 5. Pivot Tables

Create at least 3 pivot tables:

- Region-wise total sales
- Category-wise quantity sold
- Monthly sales trend (using Date field)

🔹 6. Charts

Create visualizations:

- Bar chart: Total sales by region
- Pie chart: Revenue by category
- Line chart: Monthly revenue trend

🔹 7. Final Dashboard (Optional)

Create a final Excel sheet showing:

- Summary KPIs: Total revenue, top region, top product
- Linked charts and pivot tables
- Clean, printable report format

Deliverables:

- `sales_data_cleaned.xlsx` (cleaned and formatted version)
- Pivot tables and charts in additional sheets
- Dashboard (optional but encouraged)